Final Project

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10/18/2026

121690-MIS-659-MS1 • Relational Database Management Graduate Online Fall 2026-2027

Assignment Instructions

A relational database is to be designed for a medium sized Company dealing with industrial applications of computers. The Company delivers various products to its customers ranging from a single application program through to complete installation of hardware with customized software. The Company employs various experts, consultants and supporting staff. All personnel are employed on long- term basis, i.e. there is no short-term or temporary staff. Although the Company is somehow structured for administrative purposes (that is, it is divided into departments headed by department managers) all projects are carried out in an inter-disciplinary way. For each project a project team is selected, grouping employees from different departments, and a Project Manager (also an employee of the Company) is appointed who is entirely and exclusively responsible for the control of the project, quite independently of the Company's hierarchy.

The following is a brief statement of some facts and policies adopted by the Company.

1. Each employee works in some department.
2. An employee may possess a number of skills
3. Every manager (including the MD) is an employee
4. A department may participate in none/one/many projects.
5. At least one department participates in a project.
6. An employee may be engaged in none/one/many projects
7. A team consist of at least one members

For the above business stories you are expected to:
1. Create the ERD diagram (entities, relationships with cardinality, relationship names, primary key listed for each entity).
3. Create tables in SQL and insert a minimum of 6 records in each table. Each table must contain a primary key. At least one table must contain a Foreign key.
4. Perform the Select operation for all tables
5. Demonstrate one example of Update query
6. Demonstrate one example of Alter query
7. Demonstrate one example of Join query

Submission Instructions:
Submit all of the above with SQL screenshots in One PDF document.
1. ER Diagram
2. Screenshots of SQL code in Oracle, with the results of the sql queries for all the SQL statements above.

Figure 1. Entity–Relationship (ER) Diagram

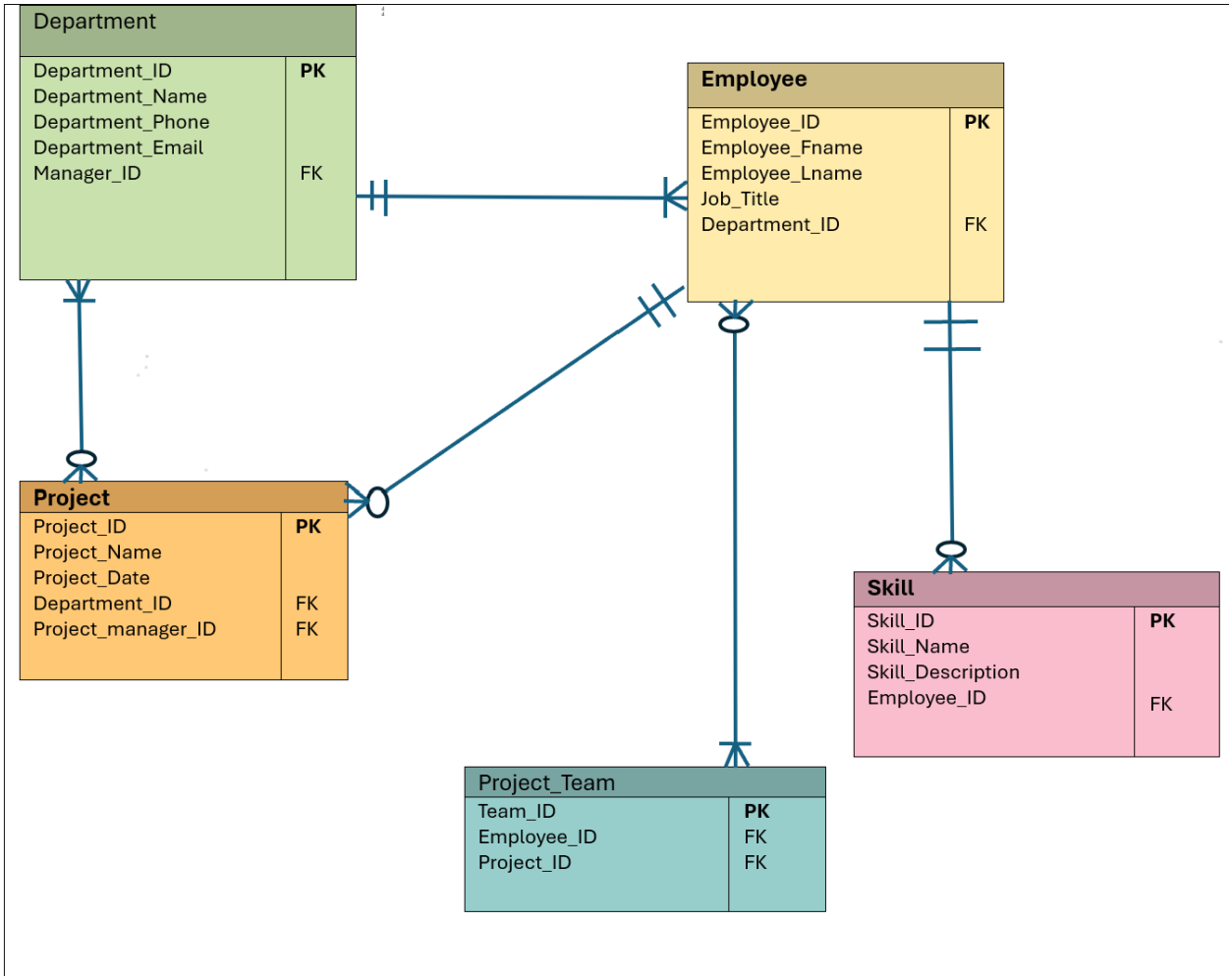


Figure 2. Creating Schema

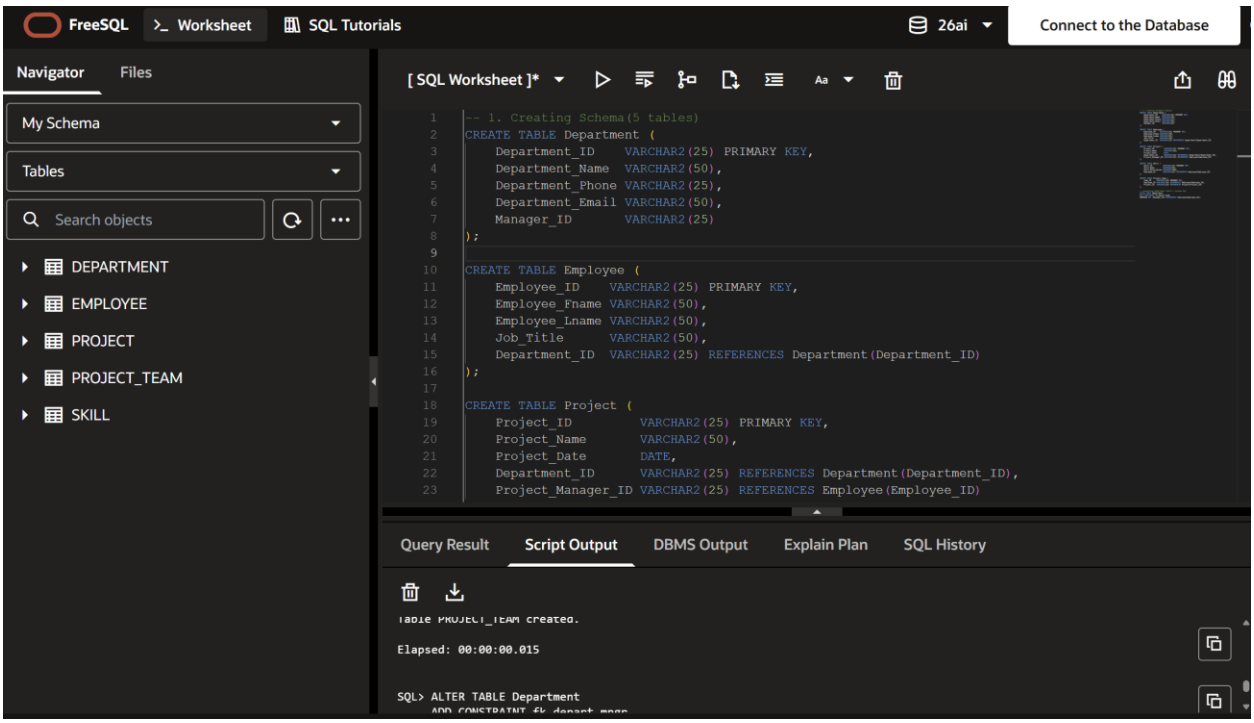


Figure 3. Finishing Schema and Using an Example of the ALTER TABLE Command

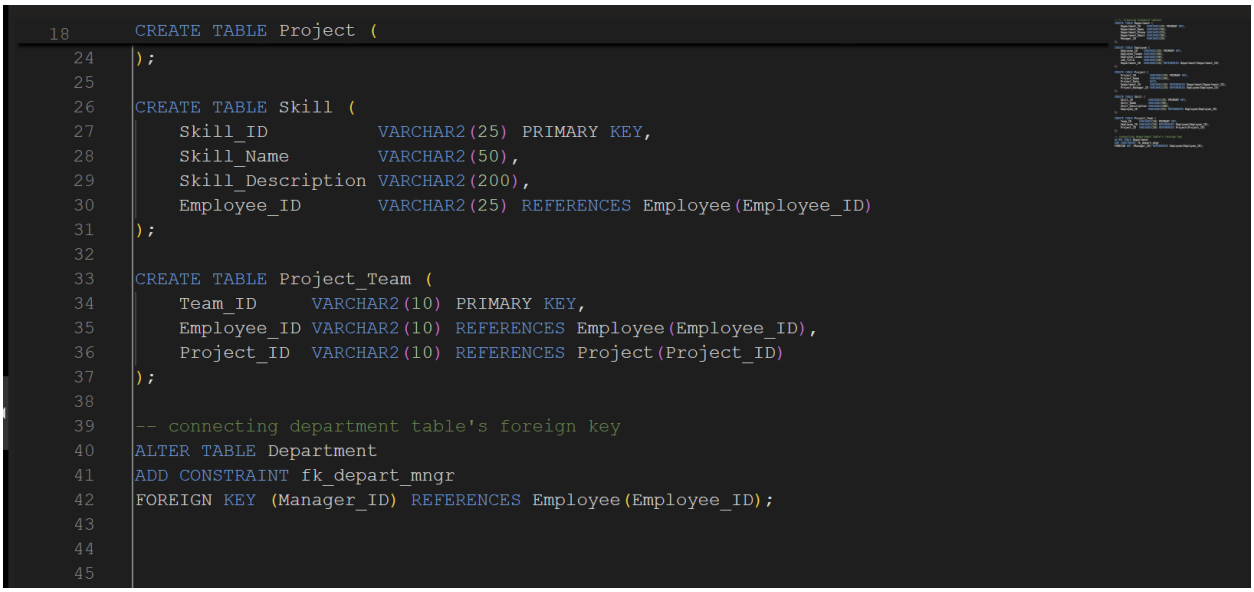


Figure 4. Inserting 6 Records

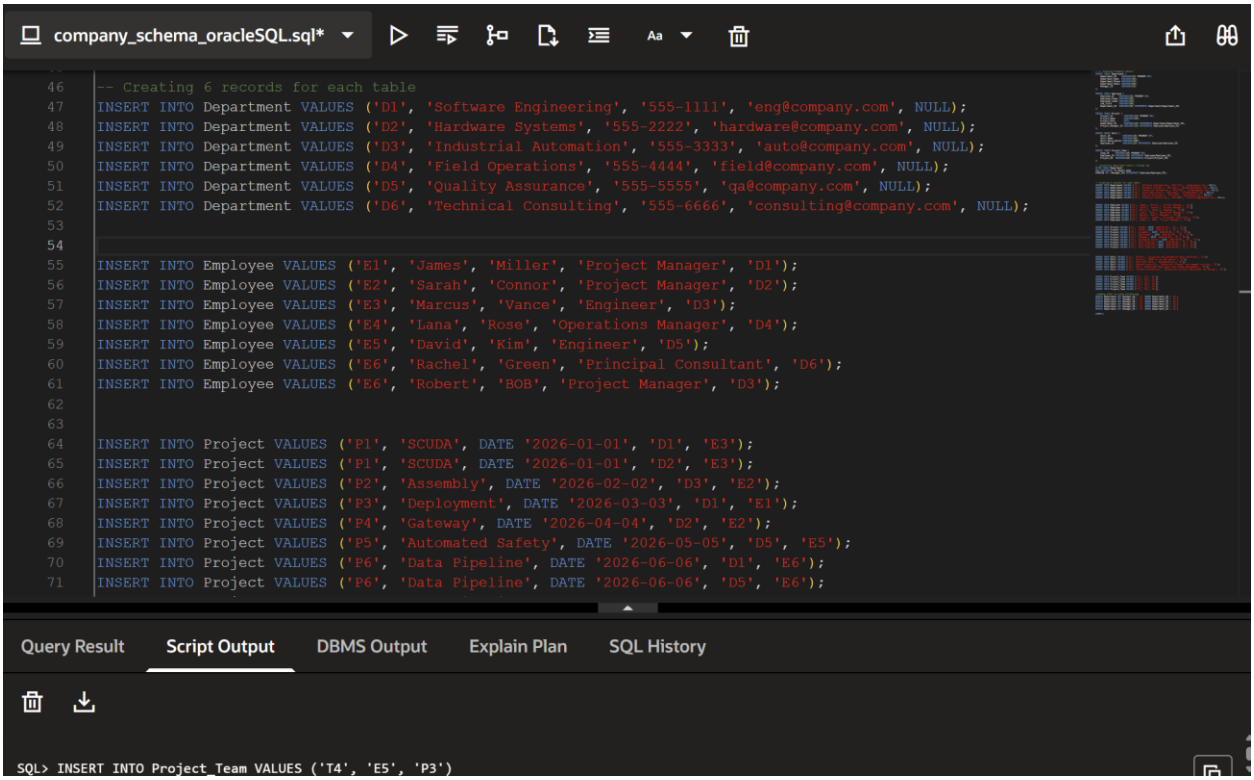
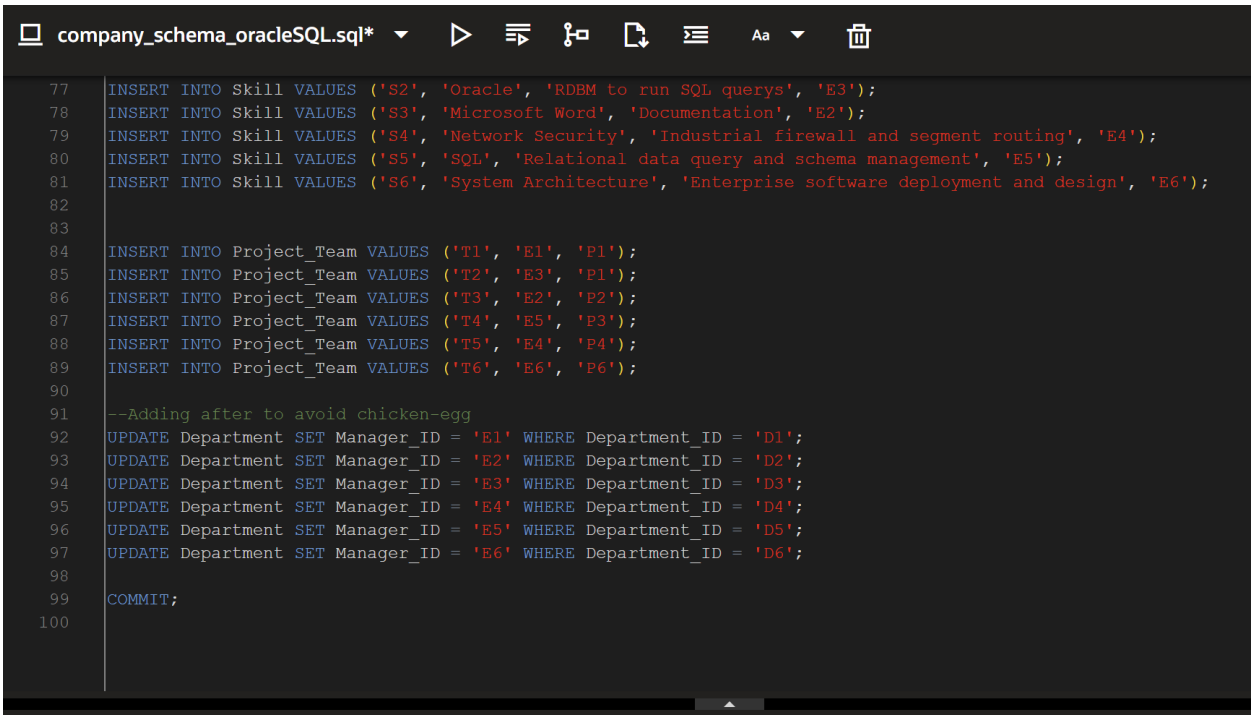
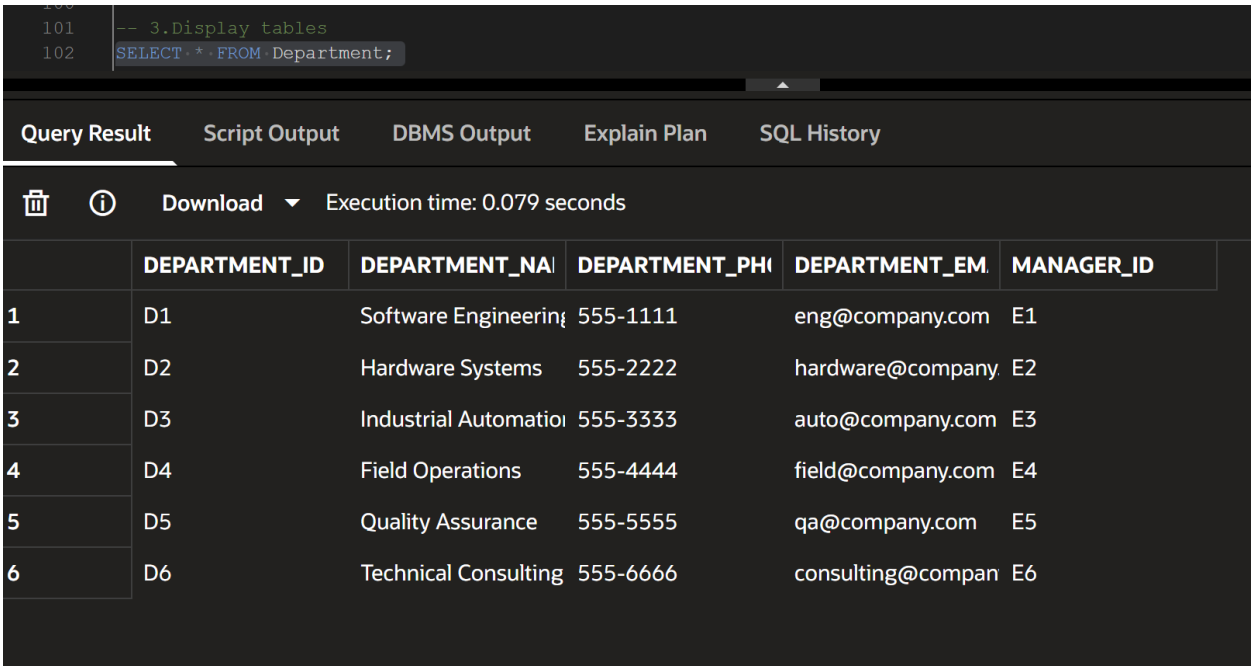


Figure 5. Completing Inserts and Utilizing the UPDATE Command



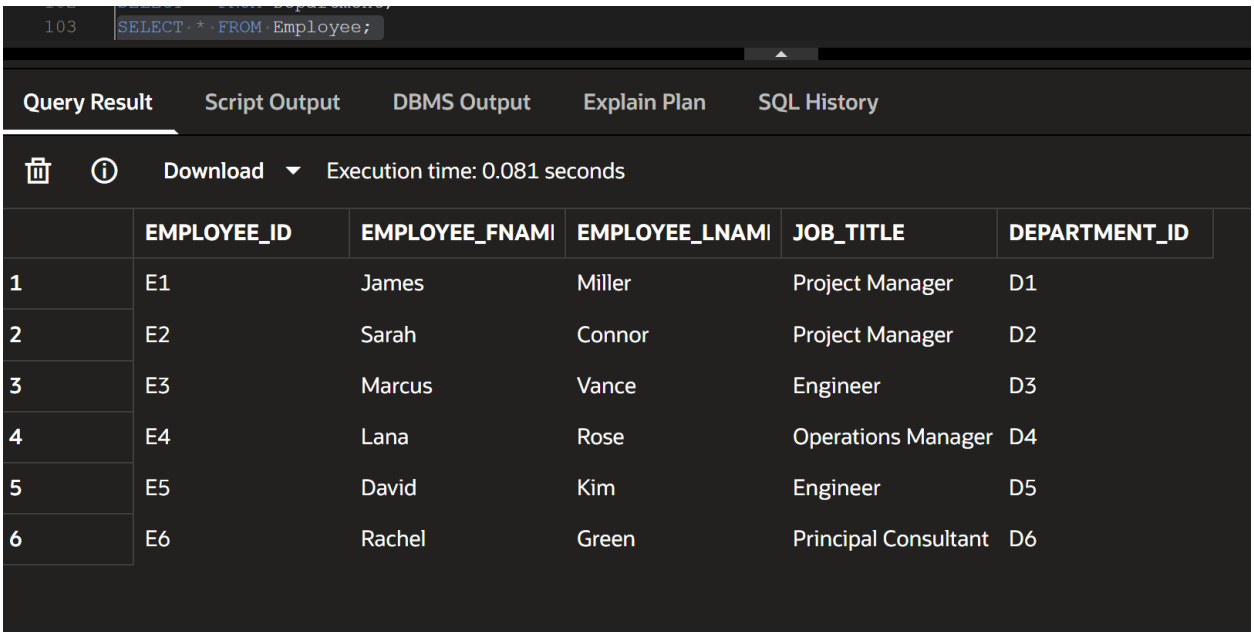
Figures 6 -10 Display Tables Utilizing SELECT

Figure 6.



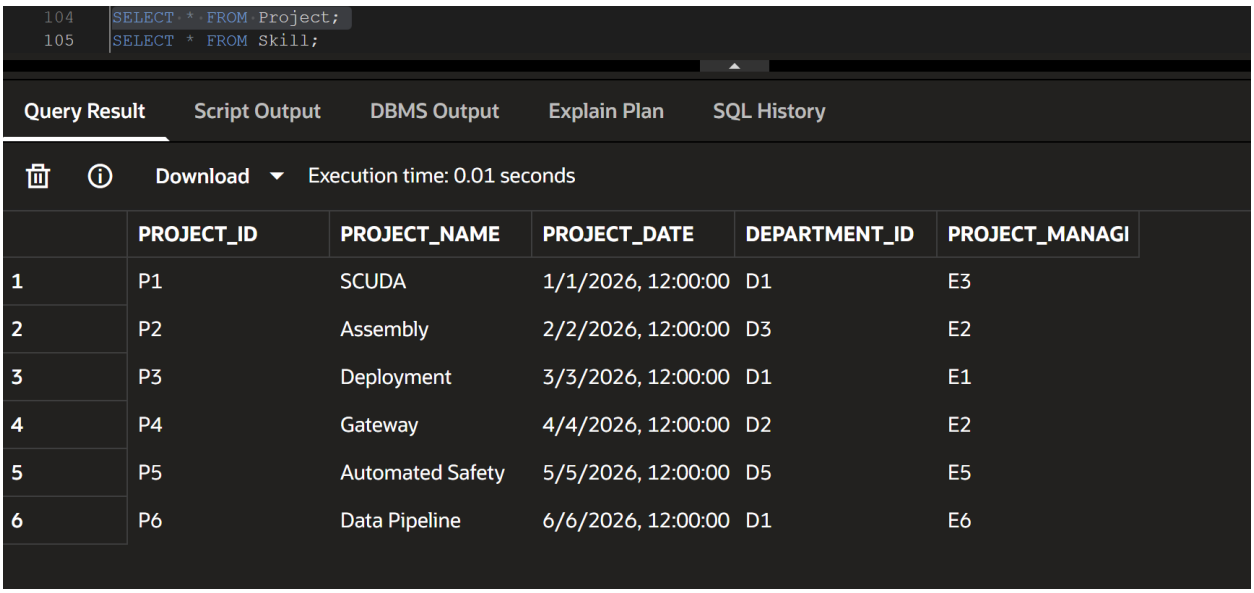
	DEPARTMENT_ID	DEPARTMENT_NAI	DEPARTMENT_PH	DEPARTMENT_EM	MANAGER_ID
1	D1	Software Engineerin	555-1111	eng@company.com	E1
2	D2	Hardware Systems	555-2222	hardware@company	E2
3	D3	Industrial Automatio	555-3333	auto@company.com	E3
4	D4	Field Operations	555-4444	field@company.com	E4
5	D5	Quality Assurance	555-5555	qa@company.com	E5
6	D6	Technical Consulting	555-6666	consulting@compan	E6

Figure 7.



	EMPLOYEE_ID	EMPLOYEE_FNAMI	EMPLOYEE_LNAMI	JOB_TITLE	DEPARTMENT_ID
1	E1	James	Miller	Project Manager	D1
2	E2	Sarah	Connor	Project Manager	D2
3	E3	Marcus	Vance	Engineer	D3
4	E4	Lana	Rose	Operations Manager	D4
5	E5	David	Kim	Engineer	D5
6	E6	Rachel	Green	Principal Consultant	D6

Figure 8.



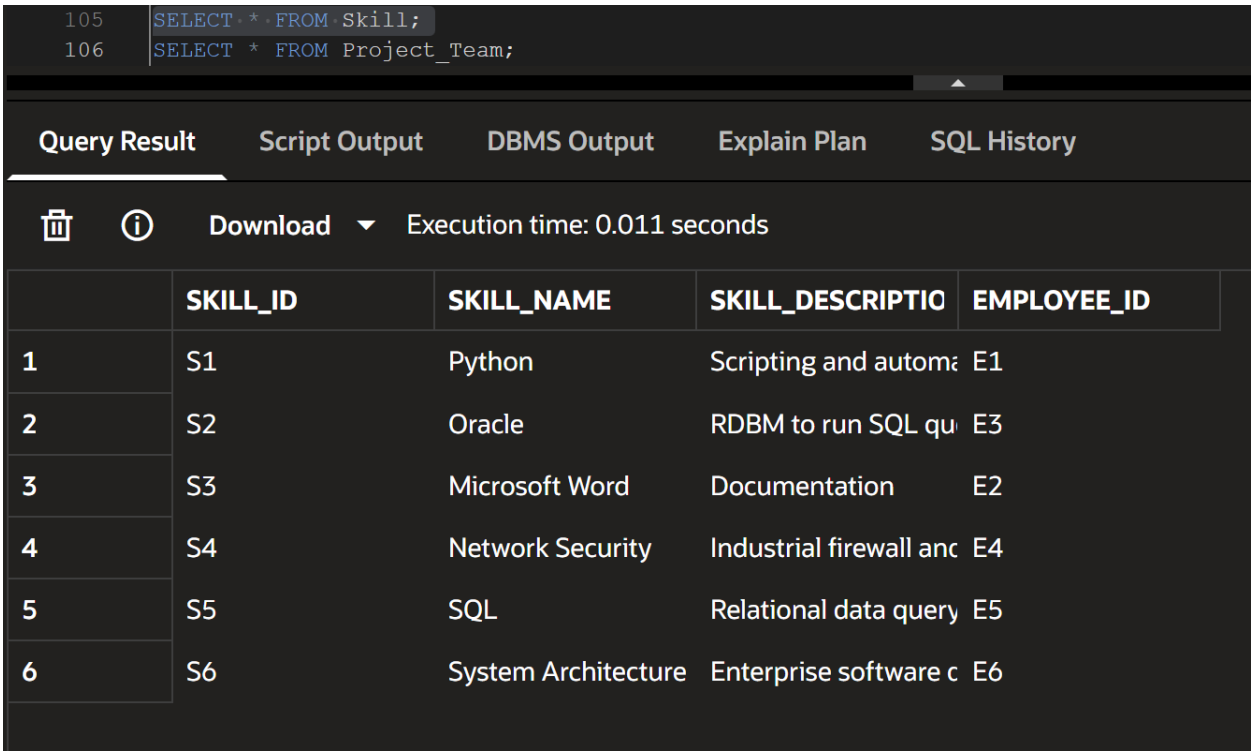
```
104  SELECT * FROM Project;
105  SELECT * FROM Skill;
```

Query Result | Script Output | DBMS Output | Explain Plan | SQL History

Download | Execution time: 0.01 seconds

	PROJECT_ID	PROJECT_NAME	PROJECT_DATE	DEPARTMENT_ID	PROJECT_MANAGI
1	P1	SCUDA	1/1/2026, 12:00:00	D1	E3
2	P2	Assembly	2/2/2026, 12:00:00	D3	E2
3	P3	Deployment	3/3/2026, 12:00:00	D1	E1
4	P4	Gateway	4/4/2026, 12:00:00	D2	E2
5	P5	Automated Safety	5/5/2026, 12:00:00	D5	E5
6	P6	Data Pipeline	6/6/2026, 12:00:00	D1	E6

Figure 9.



```
105  SELECT * FROM Skill;
106  SELECT * FROM Project_Team;
```

Query Result | Script Output | DBMS Output | Explain Plan | SQL History

Download | Execution time: 0.011 seconds

	SKILL_ID	SKILL_NAME	SKILL_DESCRIPTIO	EMPLOYEE_ID
1	S1	Python	Scripting and autom	E1
2	S2	Oracle	RDBM to run SQL qu	E3
3	S3	Microsoft Word	Documentation	E2
4	S4	Network Security	Industrial firewall anc	E4
5	S5	SQL	Relational data query	E5
6	S6	System Architecture	Enterprise software c	E6

Figure 10.

```
106  SELECT * FROM Project_Team;
```

Query Result | Script Output | DBMS Output | Explain Plan | SQL History

Download | Execution time: 0.011 seconds

	TEAM_ID	EMPLOYEE_ID	PROJECT_ID
1	T1	E1	P1
2	T2	E3	P1
3	T3	E2	P2
4	T4	E5	P3
5	T5	E4	P4
6	T6	E6	P6

Figure 11. JOINS

```
109  --4. Utilize JOINS
110  SELECT
111      Employee.Employee_ID,
112      Employee.Employee_Fname,
113      Employee.Job_Title,
114      Department.Department_Name
115  FROM Employee
116  JOIN Department
117      ON Employee.Department_ID = Department.Department_ID;
```

Query Result | Script Output | DBMS Output | Explain Plan | SQL History

Download | Execution time: 0.014 seconds

	EMPLOYEE_ID	EMPLOYEE_FNAM	JOB_TITLE	DEPARTMENT_NA
1	E1	James	Project Manager	Software Engineerin
2	E2	Sarah	Project Manager	Hardware Systems
3	E3	Marcus	Engineer	Industrial Automatio
4	E4	Lana	Operations Manager	Field Operations
5	E5	David	Engineer	Quality Assurance
6	E6	Rachel	Principal Consultant	Technical Consulting